Adventist University of Central Africa

Multi-Location Inventory Reconciliation System
CASE STUDY: B Special Business Ltd

A final year project presented in partial fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major in

Software Engineering

By

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January, 2026

ABSTRACT

A Final Project for the Bachelor's Degree in Information Technology Emphasis in Software Engineering Adventist University of Central Africa

Title: StockSync: Multi-Location Inventory Reconciliation System
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Date Completed: ..../..../2026

In emerging economies, retail expansion necessitates digital transformation in supply chains. Currently, Small and Medium Enterprises (SMEs) rely heavily on decentralized, manual inventory systems using physical notebooks and spreadsheets. This inefficiency causes critical data redundancy, recurring stock discrepancies, and delayed decision-making due to a lack of real-time visibility.

This project presents StockSync, a multi-location inventory reconciliation system for SMEs. StockSync centralizes warehouse and retail data into a unified platform. It streamlines operations through highly efficient digital data entry, automated reconciliation algorithms to instantly detect stock anomalies, and predictive analytics to forecast inventory depletion.

[Paragraph3&4 (findings and conclusion) => AFTER CHAPTER 4 As discussed]

DECLARATION

I, Adeline Tuyizere, declare that this "StockSync: Multi-Location Inventory Reconciliation System for B Special Business Ltd" project is the fruit of my effort and has not received any previous credit at the Adventist University of Central Africa or any other university or institution.

Signature: .....

Date: ...../...../2026

APPROVAL

I, Rene Ruganji, certify that this project report has been done under my supervision and submitted with my approval.

Signature: .....

Date: ...../...../2026

DEDICATION

With great pleasure, I dedicate this Research Project

To Almighty God,

To my lovely parents,

To all my family members,

To all my friends and colleagues,

To my supervisor for his kind guidance.

ACKNOWLEDGEMENTS

First and foremost, I thank God Almighty, the Source of all knowledge, wisdom, and strength, without whose grace and blessings this work would not have been possible. The completion of this project would not have been achievable without the support and guidance of many individuals to whom I owe my deepest gratitude.

I am profoundly grateful to my supervisor, Rene Ruganji, whose mentorship and thoughtful guidance played a vital role in shaping this research. His constructive feedback, patience, and academic rigor challenged me to produce work of greater depth and quality than I could have achieved alone. His dedication to my academic growth has been invaluable.

I extend my sincere appreciation to the dedicated faculty and academic staff of the Department of Software Engineering at the Adventist University of Central Africa (AUCA). Your commitment to imparting knowledge with clarity and passion, and your willingness to support students beyond the classroom, have profoundly impacted my academic and personal development. The AUCA community, guided by values of integrity, excellence, and service, has shaped not only my technical knowledge but also my character and worldview.

To my beloved parents, I am who I am because of you. Thank you for your unwavering love, prayers, financial support, and words of encouragement that sustained me through every challenge. To my siblings, thank you for your constant belief in me.

I am deeply grateful to my friends and classmates who made my university journey not only bearable but highly meaningful. Navigating the complexities of software engineering requires a strong support system, and you made the process easier. Your friendship and support have been a tremendous source of strength.

May God bless you all abundantly.

Adeline Tuyizere

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LIST OF ABBREVIATIONS

ACID: Atomicity, Consistency, Isolation, Durability

API: Application Programming Interface

AUCA: Adventist University of Central Africa

CDN: Content Delivery Network

CRM: Customer Relationship Management

CSS: Cascading Style Sheet

DB: Database

ERP: Enterprise Resource Planning

GRN: Goods Received Note

HTML: Hypertext Markup Language

JWT: JSON Web Token

MoSCoW: Must have, Should have, Could have, Won't have

MVC: Model-View-Controller

PIECES: Performance, Information, Economics, Control, Efficiency, Service

POS: Point of Sale

QR: Quick Response

RBAC: Role-Based Access Control

RDBMS: Relational Database Management System

REST: Representational State Transfer

RRA: Rwanda Revenue Authority

SME: Small and Medium Enterprises

TIN: Tax Identification Number

TLS/SSL: Transport Layer Security / Secure Sockets Layer

UAT: User Acceptance Testing

UML: Unified Modeling Language

CHAPTER ONE: GENERAL INTRODUCTION

Introduction

Rwanda's Vision 2050 and Smart Rwanda Master Plan position digital transformation as a fundamental pillar of economic competitiveness, prioritizing technology adoption across all sectors (Innovation, 2020). Within this evolving landscape, small and medium enterprises (SMEs) in distribution and supply chain sectors face mounting pressure to modernize inventory management systems to meet contemporary operational demands.

SMEs constitute Rwanda's economic backbone, yet encounter unique challenges distinguishing them from larger corporations, particularly regarding resource constraints, technological capacity, and organizational complexity (Haddara & Zach, 2011). These limitations prove especially acute in inventory management, where inability to invest in expensive enterprise resource planning (ERP) systems forces continued dependence on manual processes.

Research demonstrates that manual inventory management generates error rates of 15-20%, resulting in significant financial losses and operational inefficiencies (Odasco, 2023). When SMEs expand from single-location to multi-site operations without upgrading inventory infrastructure, they encounter compounding inefficiencies at each level. The absence of real time synchronization creates fundamental disconnects between physical stock movements and recorded data, undermining decision-making and exposing organizations to preventable losses.

This project proposes StockSync, a Multi-Location Inventory Reconciliation System addressing these critical deficiencies. By implementing automated reconciliation algorithms, real-time tracking, and predictive analytics, StockSync transforms operational efficiency while providing a scalable model for SMEs navigating Rwanda's digital transformation.

Background of the Study

SMEs in Supply Chain Management

Small and medium enterprises operate under fundamentally different constraints than large corporations. (Haddara & Zach, 2011) establish that SMEs possess distinct characteristics including limited financial resources, restricted technical expertise, and simplified organizational structures. These factors create a paradox: SMEs require efficient inventory systems to compete, yet lack capital for sophisticated ERP solutions deployed by larger competitors.

Multi-echelon supply chains further complicate inventory management. Distribution operations typically coordinate across central warehouses, regional points, and retail outlets. Sbai's (Sbai & Berrado, 2023) review demonstrates that effective multi-level coordination requires sophisticated tracking, real-time synchronization, and automated reconciliation features manual systems cannot provide.

When SMEs scale across locations without infrastructure upgrades, complexity increases exponentially beyond manual process capacity. Research in developing economies reveals persistent challenges with stock accuracy and data integrity even with basic digital tools. (Odasco, 2023) found institutions using elementary systems still struggle with discrepancies when reconciliation remains manual and periodic rather than automated and continuous. This intensifies in commercial distribution where transaction volumes and product diversity exceed institutional contexts.

Current Inventory Management Technology

Technology evolution has created a two-tiered market: sophisticated enterprise systems for large corporations versus basic spreadsheet tools inadequate for growing multi-location operations. Advanced platforms like Oracle Cloud WMS and SAP demonstrate technical feasibility of real time tracking and automated reconciliation (Sola, 2024), but implementation costs, complexity, and licensing fees exceed typical SME budgets in developing economies.

Critically, (Sudarmi & Sunaryo, 2024) confirm that even modern ERPs improve accuracy through digitization, but "reconciliation and discrepancy handling rely on operational procedures rather

than automated detection mechanisms" (p. 77). This means organizations investing in basic ERPs experience the same reconciliation challenges as manual systems merely with digitized rather than paper records.

This creates what (Haddara & Zach, 2011) identify as the "missing middle" a market void leaving SMEs without appropriate solutions. The cost structure proves particularly prohibitive: "ERP software cost as a percentage of overall cost is higher for SMEs than large enterprises" (p. 113), forcing continued operational inefficiency or expensive customization.

To bridge the gap between basic spreadsheets and expensive enterprise software, custom automation frameworks offer an effective middle ground. Research by (Omisola, 2024) demonstrates that automating key inventory workflows successfully reduces operational waste without requiring complete ERP implementation. Similarly, (Hasibuan, 2024) proved that developing a targeted, real-time tracking system across multiple locations gives SMEs exactly the functionality they need while remaining financially accessible.

The Manual Process Challenge

Traditional SME inventory management follows fragmented manual workflows creating multiple failure points. This standard manual workflow exhibits critical deficiencies. First, multiple manual transcription steps (paper → ledger → Excel → consolidation) introduce compounding errors at each transfer. Research indicates manual entry errors of 1-5% per transaction (Odasco, 2023) across thousands of monthly transactions, these create substantial discrepancies difficult to trace retrospectively.

Second, lack of real-time synchronization means discrepancies remain undetected for weeks. By monthly reconciliation, causal transactions have passed, witnesses forgot details, and evidence degraded. This temporal disconnect between occurrence and detection fundamentally undermines corrective action.

Third, absence of automated pattern detection prevents identifying systematic issues like recurring shrinkage at specific locations, shifts, or product categories. Manual reconciliation reveals only aggregate discrepancies ("Shop A short 47 units") without analytical insights. (Sudarmi &

Sunaryo, 2024) characterize this as "operational procedures" versus "systematic analytical processes" a limitation manual diligence cannot overcome.

Technological Solutions and Implementation

Modern systems leverage key technologies overcoming manual limitations. Real-time tracking via scanning and mobile entry eliminates transcription errors while providing immediate visibility (Hasibuan, 2024). Cloud-based databases ensure multi-location synchronization without complex on-premise infrastructure, eliminating needs for dedicated IT staff, server maintenance, and backup management while providing growth scalability. (Hasibuan, 2024) demonstrate cloud systems achieve real-time synchronization in seconds versus days or weeks for manual consolidation.

Automated reconciliation represents perhaps the most critical advancement. Rather than monthly audits, systems continuously compare expected stock (from receipts, transfers, sales) against actual quantities, immediately flagging discrepancies (Sola, 2024) This shift from periodic reactive detection to continuous proactive monitoring fundamentally changes inventory control dynamics.

Predictive analytics further enhance value. Analyzing historical sales velocity, seasonal trends, and depletion rates enables stock-out forecasting and proactive restocking alerts (Omisola, 2024), transforming inventory management from reactive recording to strategic planning supporting growth and competitive advantage. The V-Model development approach, successfully applied by (Hasibuan, 2024), provides structured validation at each development stage, reducing implementation risks and increasing adoption likelihood in SME contexts with limited technical expertise.

Statement of the Problem

Small and medium enterprises in Rwanda's import and distribution supply chains face critical inefficiencies from reliance on manual, non-integrated inventory management systems. Manual processes generate 15-20% error rates, causing substantial financial losses, operational delays, and compromised customer satisfaction (Odasco, 2023). These inefficiencies intensify as businesses

scale from single-location to multi-site networks, where absence of real-time synchronization creates fundamental disconnects between physical stock and recorded data.

The primary problem is absence of unified, real-time inventory tracking systems synchronizing stock data across all locations. Warehouse records, retail logs, and sales transactions exist in separate notebooks and isolated files with no automated linkage. No single person or system possesses accurate, current total inventory view. Daily operations depend on outdated, frequently incorrect information, forcing time-consuming physical verification and manual reconciliation.

Operational Impact and Manifestations

Distribution SMEs managing high-volume flows from international suppliers through central warehouses to retail outlets find paper-based systems adequate for single-store operations but wholly inadequate for multi-location complexity. Each location operates as independent data silo, with paper documents requiring manual re-entry at each step slow, error-prone, and completely unscalable.

Financial losses occur through "phantom inventory" where records and physical stock mismatch, causing lost sales and unnecessary procurement. Stock shrinkage from theft, damage, or misplacement goes undetected for weeks, with inventory shrinkage typically representing 1.5-2% annual revenue higher with poor tracking (Dennis, Wixom, & Roth, 2015).

Operational inefficiency consumes resources as managers dedicate a significant portion of their workweeks to manual counts and data entry instead of strategic activities. Sales staff waste time physically verifying availability, delaying service and reducing productivity. Compromised decision-making results from absent timely, accurate data. Procurement operates reactively on estimation rather than analysis, causing stock-outs of popular items and overstocking slow-movers.

Management lacks real-time visibility into performance, trends, or patterns, hindering market responsiveness. Competitive disadvantage emerges as rivals adopt digital tools for superior experiences. SMEs unable to guarantee availability or provide efficient service risk losing customer trust and market share. Increased operational risk stems from absent security controls, audit trails, and fraud prevention in paper systems. Lost or altered records prove irrecoverable, and

discrepancies become untraceable, exposing organizations to internal fraud and external compliance risks.

Choice and Motivation of the Study

This project emerged from observing a widening gap in Rwanda between traditional SME operations and modern digital expectations. Many distribution enterprises still rely heavily on manual processes, which causes operational inefficiencies and restricts their ability to grow. In a market where digital accessibility has shifted from being a competitive advantage to an absolute necessity, these manual limitations severely threaten a company's ability to survive and compete.

Recognizing technology could bridge this gap and catalyze growth for indigenous businesses, developing a comprehensive inventory reconciliation solution addresses authentic challenges across Rwanda's SME distribution sector, potentially benefiting numerous enterprises beyond single organizations and contributing to broader sectoral transformation.

This aligns with Adventist University of Central Africa's philosophy of cultivating graduates leveraging technical expertise to resolve substantive societal challenges. AUCA's Software Engineering curriculum emphasizes practical application of technological solutions advancing socioeconomic progress. This project manifests the university's vision of nurturing innovation serving public interests and contributing to national development objectives.

Critically, the focus on reconciliation not just recording addresses a gap where even modern ERPs rely on "operational procedures rather than automated detection mechanisms" (Sudarmi & Sunaryo, 2024).

Objectives of the Study

General Objective

To design, develop, and implement a centralized inventory reconciliation system ensuring accurate, real-time tracking and synchronization across multiple warehouse and retail locations for SMEs distribution operations, eliminating manual errors, reducing inefficiencies, preventing

financial losses from discrepancies, and establishing scalable digital foundations supporting business expansion within Rwanda's competitive retail sector.

Specific Objectives

To ensure the successful development of the system, the following SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) objectives have been established:

- Automate inventory tracking across warehouse and outlets using optimized digital search interfaces and stock registration, ensuring every stock movement is digitally recorded in real-time, eliminating manual entry errors.
- Develop automated reconciliation module continuously comparing sales against inventory, immediately flagging discrepancies, replacing monthly manual processes with daily automated reporting.
- Provide real-time dashboard offering complete inventory visibility, stock valuation, and performance metrics across locations, enabling data-driven decisions without manual compilation.
- Implement predictive analytics analyzing historical trends and product velocity to forecast depletion and generate proactive restocking alerts.
- Create a secure customer database with TIN number storage for RRA compliance record-keeping, automating audit trails and reducing legal/financial risk.
- To establish role-based access controls protecting sensitive data, maintaining encrypted audit logs, ensuring appropriate permissions across roles.

Scope of the Project

This project encompasses systematic design, development, and implementation of integrated Multi-Location Inventory Reconciliation for SME distribution enterprises, facilitating seamless warehouse, retail, and management interactions through optimized processing, enhanced coordination, and evidence-based decision-making.

Functional Coverage

- **Inventory & Transaction Tracking:** Comprehensive digital recording of all stock movements (receipts, transfers, sales, returns). The system generates complete audit trails with precise timestamps, user attribution, and location data to ensure operational accountability.
- **Automated Reconciliation Engine:** algorithms that continuously cross-reference expected stock against actual recorded quantities. The system immediately flags discrepancies and generates daily categorization reports based on variance type and financial impact.
- **Multi-Location Synchronization and dashboards:** Real-time data synchronization across the central warehouse and all retail outlets. This module establishes a single, unified source of truth and provides managers with real-time visual dashboards summarizing global inventory valuation.
- **Predictive Analytics:** Integration of sales trend analysis and depletion forecasting to generate smart alerts. By analyzing historical data and product velocity, the system predicts stock-outs before they occur to optimize procurement.
- **Role-Based Security and administration:** Strict access control mechanisms providing distinct, secure permissions for system administrators, warehouse managers, and retail cashiers. This ensures users only interact with data relevant to their specific operational tier.
- **Regulatory and Tax Compliance:** Standardized transaction recording that captures necessary metadata (such as TIN formatting and official receipt structures) to ensure the business's digital records remain compliant with local tax and auditing standards.

Technical Scope

- **Backend:** Node.js/Express RESTful API with secure endpoints, standard HTTP methods, comprehensive error handling, and structured responses.
- **Database:** PostgreSQL with ACID compliance, appropriate normalization, strategic denormalization for analytics, and optimized indexing.

- **Frontend:** React responsive web application with adaptive layouts, component architecture for reusability, and consistent UX.
- **Digital Point-of-Sale:** High-speed text search and catalog interfaces, supported by optional browser-based barcode scanning for rapid product lookup during sales and stock operations.
- **Security:** JWT authentication, role-based authorization, HTTPS encryption, SQL injection prevention, encrypted audit logs, password complexity, and session timeouts.
- **Scalability:** The software architecture is intrinsically modular, designed to accommodate future business expansion across multiple retail branches and warehouse locations without requiring structural changes to the core system.
- **Availability:** The system is architected for deployment on scalable infrastructure, designed to maintain high availability during standard operational business hours.

Excluded from Scope

Hardware procurement, customer-facing e-commerce, comprehensive accounting integration, advanced ERP modules (HR, payroll, supplier management), and active third-party logistics integration (architecture supports future integration).

Methodology and Techniques Used

The research framework synthesizes diverse methodologies establishing thorough comprehension of SME distribution operations, inventory deficiencies, and technological advancement requirements, combining qualitative and quantitative approaches through stakeholder consultations, workplace assessments, and documentation examination.

Research Design

Case study design focused on SME distribution operations provides appropriate depth for understanding complex processes and validating requirements within authentic contexts (Hasibuan, 2024). The approach integrates exploratory elements investigating SME inventory challenges and descriptive components documenting specific workflows and requirements.

Documentation Review

A comprehensive analysis was conducted on internal organizational records, peer-reviewed academic literature, and local regulatory requirements. This review ensured that the resulting system specifications were both technically sound and fully aligned with national business standards.

Structured Interviews

Semi-structured interviews captured diverse perspectives from management, warehouse staff, retail personnel, and administrative staff. Consistent frameworks explored workflows, frustrations, time allocation, error frequencies, functionality requirements, and transformation expectations.

Direct Observation

Primary data was gathered by observing daily operations in person. This involved shadowing employees through their workflows, timing specific manual processes, and documenting the frequency of human error. Additionally, an assessment of staff interaction with existing technology was conducted to ensure the new system would be user-friendly and appropriate for their technical skill levels.

System Requirements Analysis

The project utilized an organized synthesis process to translate raw data into technical specifications. This involved developing detailed use cases to map user interactions and applying the MoSCoW methodology to prioritize core functionalities over secondary features. Additionally, technical and operational constraints were identified and reviewed during validation sessions with stakeholders to ensure the proposed solution was both feasible and effective.

Development Methodology

The project adopts a hybrid development approach, combining the flexibility of Agile iterations with the structured validation of the V-Model, as advocated by (Hasibuan, 2024). Development is conducted in iterative two-week sprints, delivering functional software increments for regular

stakeholder review. To ensure high reliability, a continuous testing framework is employed, encompassing unit, integration, system, and user acceptance testing (UAT) to validate the system's usability and stability before final deployment.

Expected Results

System Deliverables

- **Fully Functional Web Application:** Deployed system with secure authentication operating on standard browsers.
- **Comprehensive Documentation:** Technical documentation and user documentation.
- **Source Code Repository:** Complete, well-documented code with version control enabling future maintenance.

Operational Improvements

- **Enhanced Accuracy:** Automated reconciliation substantially reduces discrepancies through continuous monitoring.
- **Real-Time Visibility:** Instantaneous updates on movements, levels, and processing with automated alerts.
- **Optimized Resources:** The system uses data to make sure the company spends its money on the right products and spends its employees' time on the right tasks.
- **Strengthened Decisions:** Continuous monitoring of patterns enables informed interventions.
- **Improved Compliance:** Customer TIN storage and automated audit trails ensure regulatory compliance.

Scalability and Future Integration

Modular architecture supports progressive integration of mobile applications, CRM systems, logistics protocols, and ERP connectivity as needs evolve. This ensures operational relevance and adaptability aligned with innovations and growth trajectories.

Organization of the Thesis

Chapter One: General Introduction provides comprehensive research context including SME supply chain background, detailed problem statement, SMART objectives, scope definition, and methodological approach.

Chapter Two: Analysis of the Current System examines existing operational frameworks, concentrating on inventory workflows, transaction processing, and data management. PIECES framework systematically evaluates limitations justifying technological intervention.

Chapter Three: Requirement Analysis and Design presents logical and technical conceptualization including theoretical frameworks, system architecture, and database specifications. UML diagrams illustrate architectural design, functional relationships, and operational logic.

Chapter Four: Implementation documents practical development and deployment including implemented functionalities with screenshots, technological frameworks, infrastructure prerequisites, validation procedures, and deployment methodology.

Chapter Five: Conclusion and Recommendations synthesizes principal findings, articulates transformative impact, and formulates recommendations for enhancement and scholarly inquiry.

CHAPTER TWO: ANALYSIS OF THE CURRENT SYSTEM

Introduction

Before engineering a comprehensive technological solution, established software engineering methodologies dictate that a rigorous, systematic evaluation of the existing operational environment must be conducted (Sommerville, 2015). This analytical phase is critical for identifying specific workflow bottlenecks, understanding user interactions, and defining the precise parameters of the operational problem space.

This chapter provides a highly detailed, comprehensive analysis of the current manual inventory and distribution system utilized by B Special Business Ltd. It begins by establishing the organizational context, including the historical background, mission, and vision of the enterprise. Subsequently, it deconstructs the "AS-IS" process model, mapping the physical and informational flow of goods across multiple echelons of the supply chain.

The system's inherent vulnerabilities are then critically evaluated using the PIECES (Performance, Information, Economics, Control, Efficiency, Service) analytical framework (Wetherbe, 2012). Finally, drawing from these documented deficiencies, the chapter establishes the foundational functional and non-functional requirements that the proposed StockSync digital architecture must fulfill to drive institutional digital transformation.

Description of Current System Environment

Historical Background

B Special Business Ltd was established in Kigali, Rwanda, to bridge a critical gap in the regional supply chain between international product manufacturers and the rapidly growing Rwandan retail market. Initiating operations as a localized, single-store trading entity, the company capitalized on rising consumer demand to steadily expand its operational footprint. Today, B Special Business Ltd has evolved into a recognized, mid-sized importer and wholesale distributor within the capital city.

The company currently operates a decentralized organizational structure to manage its high-volume operations. At the core of its supply chain is a central warehouse facility, which serves as the primary receiving hub for bulk international shipments. From this central node, inventory is systematically distributed to multiple company-owned retail branches strategically positioned across high-traffic commercial zones in Kigali.

The workforce comprises specialized personnel distributed across these nodes, including procurement officers, central warehouse managers, logistics drivers, retail branch managers, and point-of-sale cashiers. Despite its physical expansion and growing market share, the enterprise's internal data management infrastructure has not evolved proportionately, remaining heavily reliant on traditional, paper-based administrative methodologies that increasingly threaten to constrain its future scalability.

Mission

To serve as Rwanda's most trusted and reliable import and distribution partner by delivering high-quality commercial products with unmatched operational efficiency, sustained product availability, and an unwavering commitment to customer satisfaction across all retail touchpoints.

Vision

To build a modern, efficient, and transparent distribution network that guarantees consistent product availability and seamless service across all our branches.

Description of the Current System

The current Internal Inventory and Distribution Model at B Special Business Ltd operates on a strictly manual, localized data-capture paradigm. The entire supply chain data flow relies on physical documentation passing from one geographic location to another, without any centralized digital repository.

This operational workflow is categorized into four distinct chronological phases:

Phase 1: Inbound Logistics and Warehouse Reception. The inventory cycle begins when bulk shipments arrive at the central warehouse from international suppliers. The central Warehouse Manager is tasked with manually offloading and counting the physical goods. Upon verification of the physical count against the supplier's invoice, the manager transcribes this data into a large, physical master ledger known as the "Goods Received Note" (GRN) logbook. This physical book remains securely locked in the warehouse office and is the sole authoritative record of the company's total imported assets.

Phase 2: Intra-Company Stock Requisition and Transfer. As retail branches deplete their local stock, Branch Managers must request replenishments. Because they cannot digitally view the central warehouse inventory, this is typically done via informal communication channels. Upon receiving a request, the Warehouse Manager manually drafts a paper "Transfer Note" in duplicate. The requested goods are physically counted, loaded onto a transport vehicle, and the original paper Transfer Note is handed to the logistics driver. The warehouse ledger is manually updated to reflect the deduction.

Phase 3: Branch Reception and Point-of-Sale (POS). Upon arrival at the retail branch, the Branch Manager or Cashier must physically recount the delivered goods to ensure they match the driver's paper Transfer Note. Once verified, the items are placed on the retail shelves. The branch maintains its own separate, physical "Shop Ledger" where incoming goods are recorded. As retail customers purchase items, the cashiers manually document each transaction into a daily "Sales Log" notebook.

Phase 4: Month-End Consolidation and Auditing. Because the Central Warehouse Ledger and the diverse Retail Branch Sales Logs are entirely disconnected, B Special Business Ltd operates with fragmented data silos. Executive management does not have a real-time view of total company inventory. To calculate revenue and identify stock levels, management must wait until the end of the month. Auditors physically collect the Sales Logs from all branches, retrieve the GRN ledger from the warehouse, and spend days manually typing thousands of handwritten transactions into spreadsheets to attempt a retroactive reconciliation.

Modeling of the Current System (AS-IS Model)

To fully illustrate the functional disconnects and data silos inherent in the manual processes of B Special Business Ltd, an AS-IS Process Model has been formulated utilizing Unified Modeling Language (UML) standards for activity diagramming (Dennis, Wixom, & Roth, 2015). The diagram below visually maps the flow of physical inventory and its corresponding, disjointed paper trails across the various actors within the supply chain.

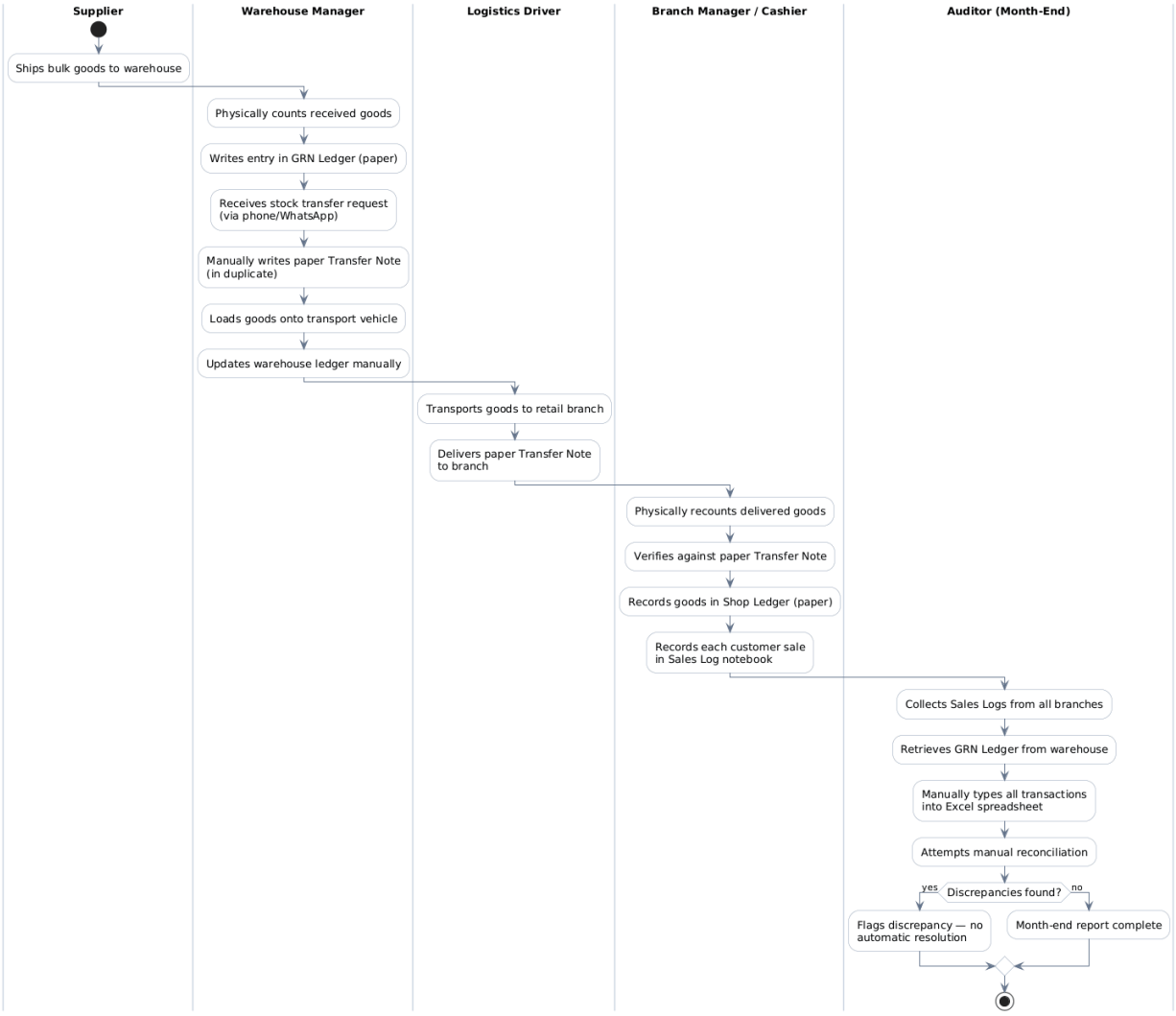

Figure 1: AS-IS Process Model and Data Flow for B Special Business Ltd

Problems with the Current System

Based on the AS-IS process model, the specific vulnerabilities of the manual inventory system are categorized below using the PIECES framework:

Performance

The system exhibits critically low throughput and unacceptable data response times. Because information moves at the speed of physical paper, executing a routine stock transfer between the warehouse and a retail branch requires hours of manual paperwork and verification. Generating comprehensive performance reports takes days of manual data aggregation across multiple branches.

Information

The integrity, accuracy, and timeliness of organizational data are fundamentally compromised. The existence of decentralized "data silos" means that warehouse ledgers and shop sales logs rarely synchronize perfectly. Discrepancies between the recorded ledger data and physical shelf counts are commonplace, driven by human transcription errors, delayed ledger entries, and arithmetic miscalculations.

Economics

The manual system inflicts direct, quantifiable financial damage on the enterprise. The inability to track stock movements in real-time creates an environment where inventory shrinkage remains completely undetected for up to four to six weeks until the next monthly audit. As supported by (Sudarmi & Sunaryo, 2024), inventory shrinkage caused by disjointed manual tracking typically represents significant annual revenue loss that cannot be recovered retrospectively.

Control

The current operational environment provides virtually no security, accountability, or verifiable audit trails. Physical notebooks and paper ledgers are inherently vulnerable; they can be easily misplaced, intentionally altered, or physically damaged without any possibility of digital recovery.

Because entries are handwritten without robust authentication, it is nearly impossible to trace accountability back to a specific employee when valuable stock goes missing.

Efficiency

The workflow is characterized by a massive duplication of effort and severe data redundancy. The exact same data variables (product name, SKU, date, quantity) are written manually by the warehouse manager upon receipt, transcribed again onto a transfer note, written again by the receiving branch manager, and finally typed into an Excel spreadsheet by an auditor.

Service

The culmination of these internal failures directly degrades the external customer experience. Because retail staff do not possess real-time visibility into central warehouse stock levels, they frequently rely on outdated knowledge to confirm product availability for wholesale buyers. This results in significant service delays and customer frustration when staff physically search the warehouse only to discover empty shelves.

Proposed Solutions

To decisively resolve the systemic vulnerabilities identified within the PIECES framework, this project proposes the design, development, and deployment of StockSync an Intelligent Multi-Location Inventory Reconciliation System. StockSync is envisioned as a comprehensive web and mobile-responsive application specifically engineered to bridge the digital gap in SME supply chains. The proposed system directly targets the root cause of B Special Business Ltd's inefficiencies: data fragmentation.

By centralizing all inventory operations into a single database, StockSync ensures that the central warehouse and all retail branches operate from a single source of truth. The application utilizes a centralized digital catalog where products are tracked via unique identifiers (SKUs). This allows retail staff to process sales instantaneously using an optimized, fast-search interface accessible from any standard web-enabled device.

Crucially, StockSync transcends basic digital record-keeping by integrating an Automated Reconciliation Engine. This engine will continuously and autonomously cross-reference daily point-of-sale retail transactions against central warehouse stock deductions, instantly flagging discrepancies at the close of each business day rather than at the end of the month.

System Requirements

The successful realization of the StockSync architecture necessitates the formal definition of rigorous system requirements. These requirements serve as the fundamental blueprint for the software engineering phase (Sommerville, 2015), ensuring the final application precisely addresses the documented operational deficiencies.

Functional Requirements

- **User Access and Role Management:** The system must allow the super-administrator to create, read, update, and deactivate user accounts with strict Role-Based Access Control (RBAC).
- **Digital Stock Registration:** The system includes a product registration module allowing warehouse staff to log incoming shipments with digital identifiers, enabling rapid database lookup at the point of sale and during stock operations.
- **Multi-Echelon Stock Transfer:** The system must facilitate digital transfer requisitions, allowing retail branches to request stock digitally.
- **Point-of-Sale (POS) Integration:** The system must provide a mobile-responsive interface allowing retail cashiers to record customer transactions, which immediately deduct stock quantities from that branch's localized database.
- **Automated Discrepancy Reconciliation:** The system must feature an autonomous computational engine that executes at the close of daily operations, cross-referencing total logged sales against physical inventory deductions to generate immediate variance reports.
- **Predictive Restocking Alerts:** The system must analyze historical sales data and current depletion rates to forecast stock-outs.

- **Compliance and Customer Data:** The system must securely store wholesale customer profiles and validate the format of Rwandan Tax Identification Numbers.

Non-Functional Requirements

- **Performance and Latency:** The system must process database write-operations and reflect inventory updates across all dashboard views with minimal latency, achieved through optimized PostgreSQL queries and indexed relational tables.
- **Security and Authentication:** All user passwords must be cryptographically hashed. System access and API endpoints must be secured utilizing JSON Web Tokens (JWT).
- **Reliability and Data Integrity:** The underlying relational database (PostgreSQL) must adhere to strict ACID compliance standards to prevent partial data corruption.
- **Scalability:** The software architecture is intrinsically modular, designed to accommodate future business expansion across multiple retail branches and warehouse locations without requiring structural changes to the core system.
- **Usability and Accessibility:** The frontend interfaces must employ responsive design principles, ensuring seamless operability across standard desktop monitors and mobile tablet devices.
- **Availability:** The system is architected for deployment on scalable infrastructure, designed to maintain high availability during standard operational business hours.

CHAPTER THREE: REQUIREMENT ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

The successful implementation of any robust software solution relies heavily on a rigorous design phase that translates theoretical business requirements into concrete technical blueprints (Pressman, 2014). Having established the functional and non-functional requirements necessary to resolve the manual inefficiencies at B Special Business Ltd in Chapter 2, this chapter focuses exclusively on the architectural and logical design of the proposed system: StockSync.

The chapter details the overarching system architecture, presents Unified Modeling Language (UML) diagrams to illustrate system behavior, outlines the comprehensive database schema through Entity-Relationship modeling, establishes the User Interface (UI) design principles, and defines the specific technology stack chosen to ensure system scalability and security.

Unified Modeling Language (UML)

To provide a standardized, globally recognized visual representation of the StockSync platform's functional logic, Unified Modeling Language (UML) diagrams have been engineered based on the standards outlined by (Fowler, 2003). These diagrams serve as the primary communication mechanism between the system designers and the development team.

Design of the New System

Use Case Diagram

The Use Case diagram defines the operational boundaries of the system and illustrates the specific functional interactions between external actors and internal processes.

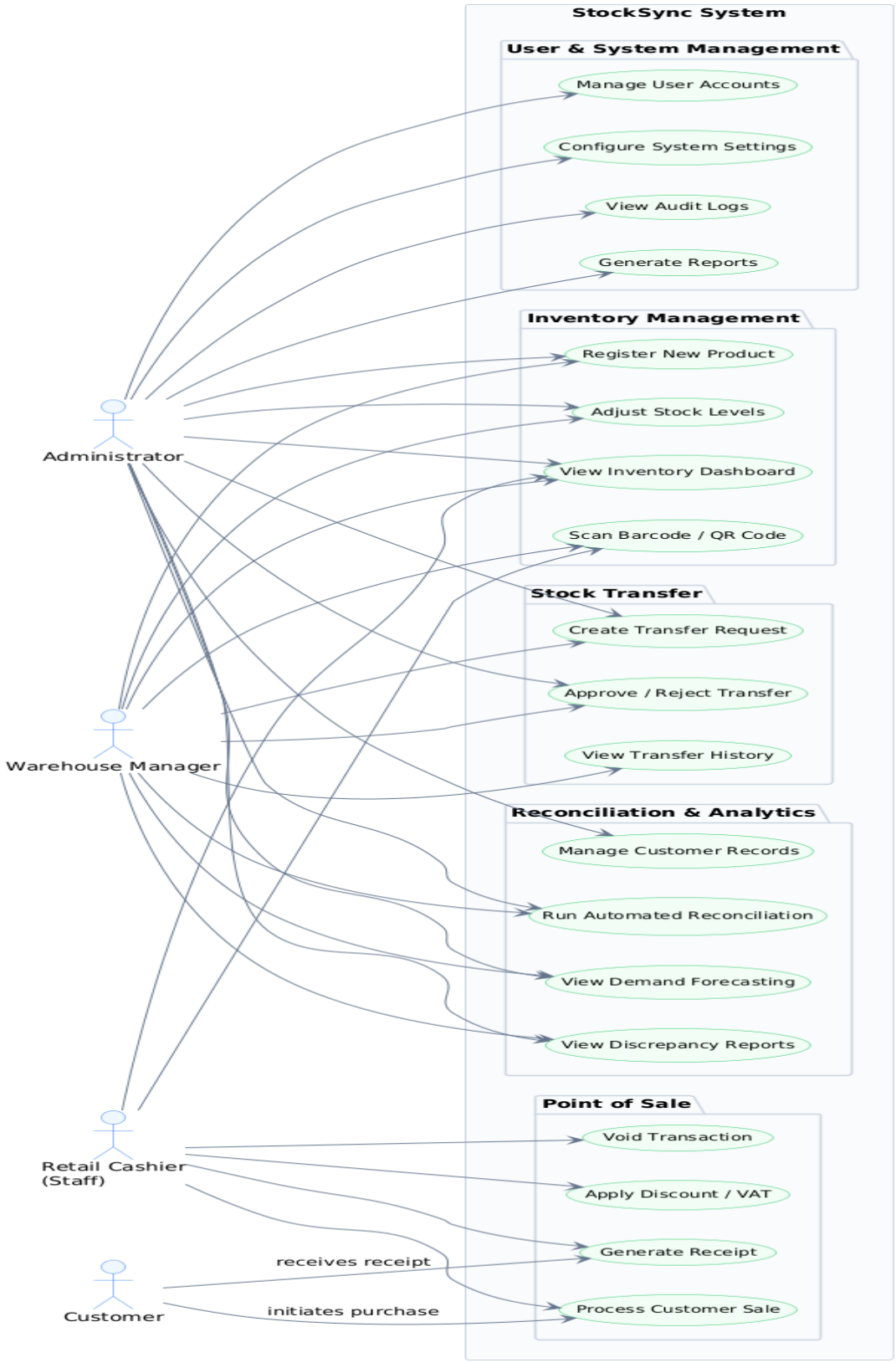

Figure 2: UML Use Case Diagram for the StockSync System

As illustrated in Figure 2, the operational privileges are strictly segregated. The Administrator holds overarching privileges, including system configuration, user management, audit log access, and reconciliation oversight. The Warehouse Manager is authorized to manage inventory, register products, and approve multi-echelon stock transfers. The Retail Cashier's interactions are intentionally restricted to the point-of-sale interface. The Customer is represented as an external actor who initiates purchases and receives receipts but does not interact directly with the system..

Sequence Diagram

While the Use Case diagram illustrates what the system does, the Sequence Diagram maps how objects interact within the system chronologically to execute a specific function.

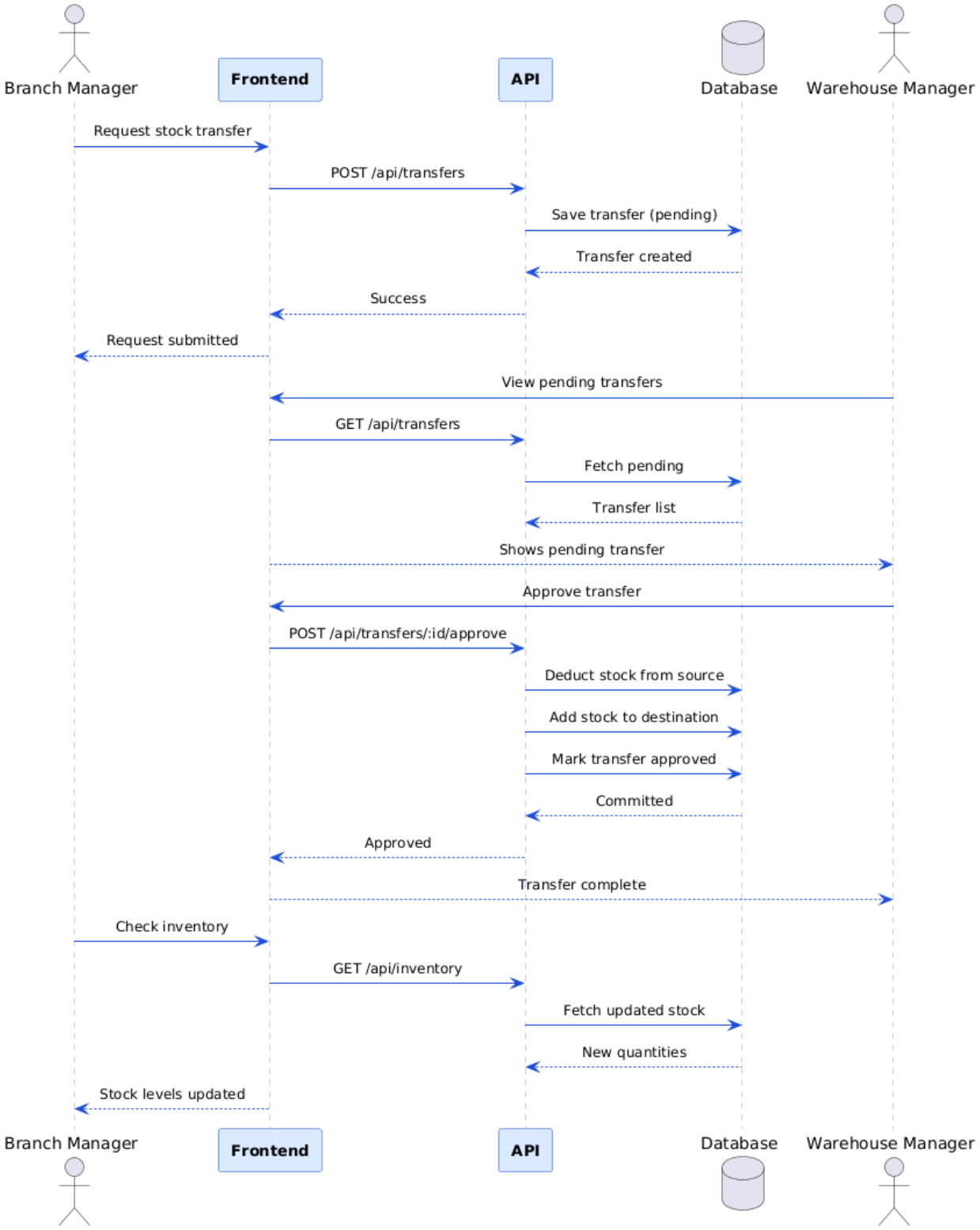

Figure 3: Sequence Diagram for the StockSync system

This sequence resolves the critical "Data Silo" problem. When a Branch Manager digitally requests stock, the Warehouse Manager initiates a transfer. The backend API updates the database, placing stock into "Transit". Upon physical arrival, the Branch Manager confirms receipt in the system, triggering a database transaction that simultaneously deducts items from the warehouse and appends them to the retail branch, ensuring ACID compliance and eliminating manual transcription errors.

Class Diagram

The Class Diagram maps the static structural blueprint of the StockSync application, serving as the direct blueprint for the Object-Oriented Programming (OOP) implementation within the backend environment.

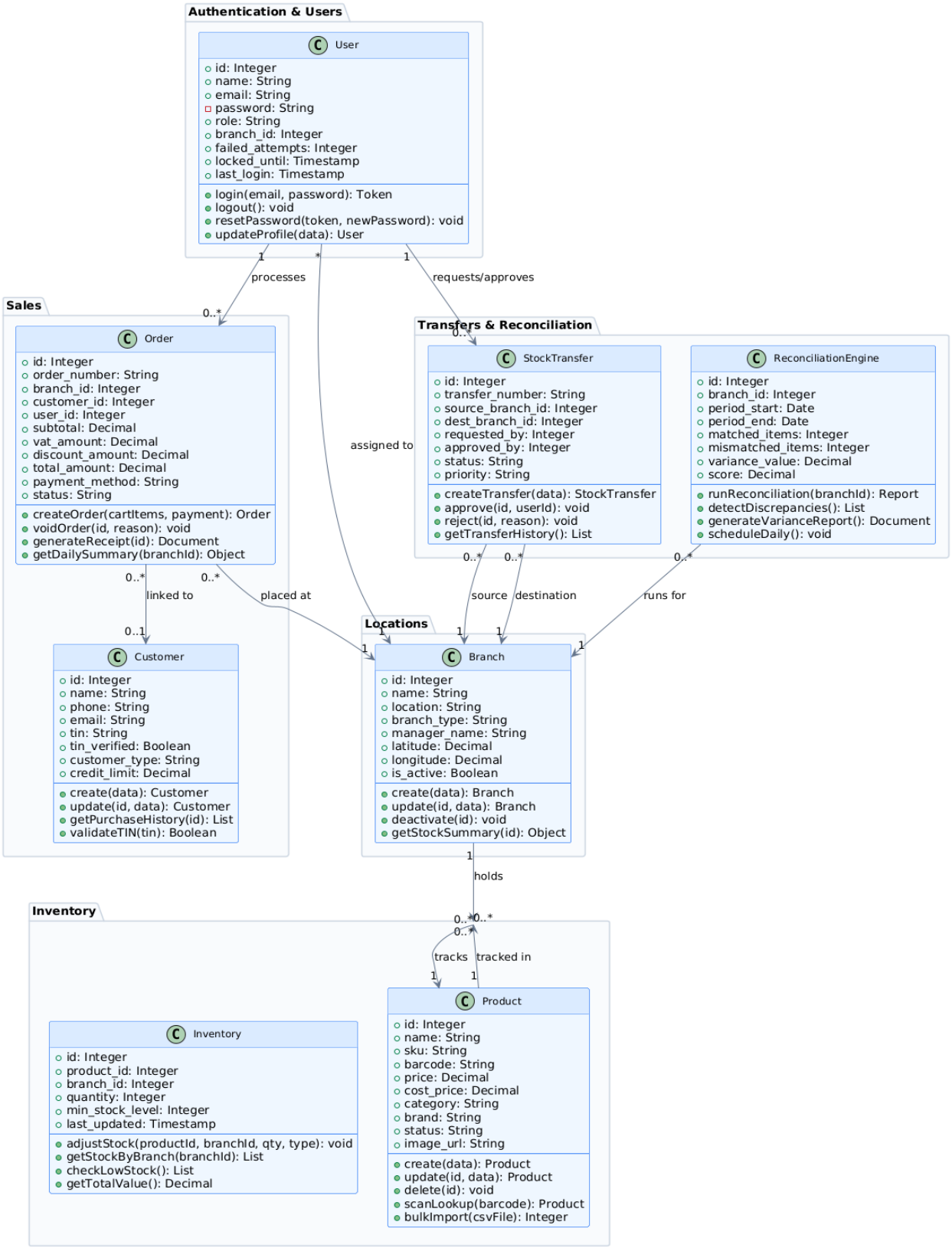

Figure 4 Class diagram for the StockSync Sytem

Database Diagram

Data integrity, rapid query performance, and transactional security are paramount. StockSync utilizes PostgreSQL, a robust Relational Database Management System (RDBMS), to handle high-concurrency transactions.

Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram provides the conceptual architecture of the data layer, defining the entities and the logical relationships that govern the database.

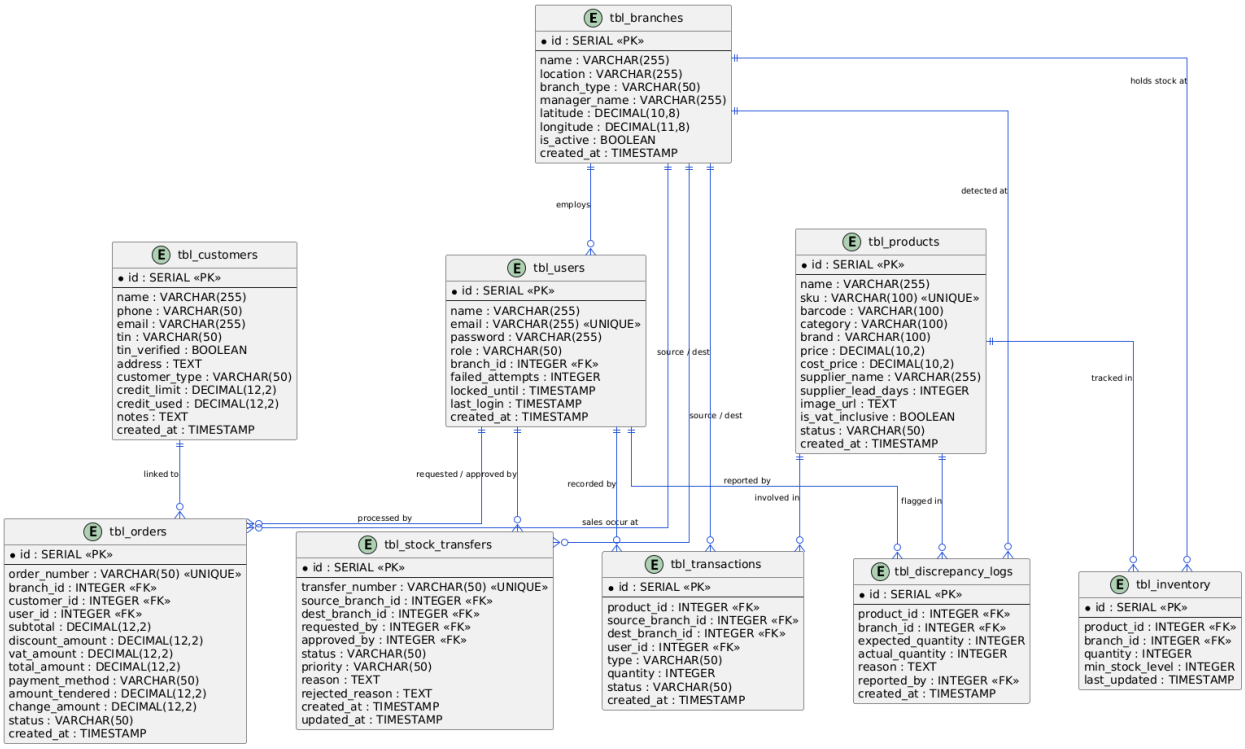

Figure 5: Entity Relationship diagram for StockSync

Comprehensive Data Dictionary

To ensure precise database instantiation, the physical schema has been strictly normalized to the Third Normal Form (3NF) to eliminate data redundancy and prevent insertion anomalies (Elmasri

& Navathe, 2015). Below are the structural definitions of the 9 core tables constituting the StockSync system.

Users Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier for each user
name	VARCHAR(255)	Not Null	Full name of the employee
email	VARCHAR(255)	Unique, Not Null	Email address used as login credential
password	VARCHAR(255)	Not Null	Bcrypt cryptographically hashed password
role	VARCHAR(50)	Default 'staff'	Access level: admin, manager, staff
branch_id	INTEGER	Foreign Key → branches(id)	Links the user to their assigned operational branch

failed_attempts	INTEGER	Default 0	Tracks consecutive failed login attempts for lockout
locked_until	TIMESTAMP	Nullable	Timestamp until which the account remains locked
last_login	TIMESTAMP	Nullable	Timestamp of the user's most recent successful login
created_at	TIMESTAMP	Default NOW()	Timestamp when the account was created

Table 1: Users Table

Branches Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier for each location
name	VARCHAR(255)	Not Null	Commercial name of the branch or warehouse

location	VARCHAR(255)	Nullable	Physical address of the branch
contact	VARCHAR(255)	Nullable	Contact information for the branch
branch_type	VARCHAR(50)	Default 'warehouse'	Classification: warehouse or retail
manager_name	VARCHAR(255)	Nullable	Name of the branch manager
latitude	DECIMAL(10,8)	Nullable	GPS latitude coordinate for geolocation mapping
longitude	DECIMAL(11,8)	Nullable	GPS longitude coordinate for geolocation mapping
is_active	BOOLEAN	Default TRUE	Indicates whether the branch is currently operational
created_at	TIMESTAMP	Default NOW()	Timestamp when the branch record was created

Table 2: Branches Table

Products Table

Attribute Name	Data Type	Constraint	Description
id	SERIAL	Primary Key	Auto-increment unique identifier for each product
name	VARCHAR(255)	Not Null	Commercial name of the product
sku	VARCHAR(100)	Unique, Not Null	Stock Keeping Unit code for inventory tracking
barcode	VARCHAR(100)	Nullable	Barcode value for scanner-based product lookup
description	TEXT	Nullable	Detailed product description
category	VARCHAR(100)	Nullable	Product category (e.g. Soft Drinks, Water, Sparkling Juice)

brand	VARCHAR(100)	Nullable	Brand or manufacturer name
price	DECIMAL(10,2)	Not Null	Standard retail selling price in RWF
cost_price	DECIMAL(10,2)	Default 0	Purchase cost from supplier in RWF
supplier_name	VARCHAR(255)	Nullable	Name of the primary supplier
supplier_lead_days	INTEGER	Default 0	Number of days required for supplier restocking
image_url	TEXT	Nullable	File path to the product image
is_vat_inclusive	BOOLEAN	Default TRUE	Indicates whether the price includes 18% VAT
status	VARCHAR(50)	Default 'active'	Product status: active or inactive
created_at	TIMESTAMP	Default NOW()	Timestamp when the product was registered

Table 3: Products Table

Inventory Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier
product_id	INTEGER	Foreign Key → products(id)	The product whose stock is being tracked
branch_id	INTEGER	Foreign Key → branches(id)	The location holding this stock
quantity	INTEGER	Default 0	Current real-time stock level at this location
min_stock_level	INTEGER	Default 10	Minimum threshold that triggers a low stock alert

last_updated	TIMESTAMP	Default NOW()	Timestamp of the most recent stock update

Table 4: Inventory Table

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Transaction Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier
product_id	INTEGER	Foreign Key → products(id)	The product involved in the movement
source_branch_id	INTEGER	Foreign Key → branches(id)	Origin branch of the stock movement
dest_branch_id	INTEGER	Foreign Key → branches(id)	Destination branch of the stock movement
type	VARCHAR(50)	Not Null	Movement type: receive, sale, transfer, or adjustment

quantity	INTEGER	Not Null	Number of units involved in the transaction
user_id	INTEGER	Foreign Key → users(id)	Employee who recorded the transaction
status	VARCHAR(50)	Default 'completed'	Transaction status
created_at	TIMESTAMP	Default NOW()	Exact timestamp of the transaction for audit trailing

Table 5:Transactions Table

Orders Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier
order_number	VARCHAR(50)	Unique, Not Null	Human-readable receipt reference number

branch_id	INTEGER	Foreign Key → branches(id)	The retail branch where the sale occurred
customer_id	INTEGER	Foreign Key → customers(id)	Optional link to a registered customer
user_id	INTEGER	Foreign Key → users(id)	Cashier who processed the sale
subtotal	DECIMAL(12,2)	Default 0	Total before VAT and discounts in RWF
discount_amount	DECIMAL(12,2)	Default 0	Total discount applied in RWF
vat_amount	DECIMAL(12,2)	Default 0	VAT amount charged (18%) in RWF
total_amount	DECIMAL(12,2)	Default 0	Final amount paid by the customer in RWF
payment_method	VARCHAR(50)	Default 'cash'	Payment method: cash, momo, or card

amount_tendered	DECIMAL(12,2)	Default 0	Amount given by the customer in RWF
change_amount	DECIMAL(12,2)	Default 0	Change returned to the customer in RWF
status	VARCHAR(50)	Default 'completed'	Order status: completed or voided
created_at	TIMESTAMP	Default NOW()	Timestamp of the sale transaction

Table 6: Orders Table

Stock Transfers Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier
transfer_number	VARCHAR(50)	Unique, Not Null	Human-readable transfer reference number

source_branch_id	INTEGER	Foreign Key → branches(id)	Branch sending the stock
dest_branch_id	INTEGER	Foreign Key → branches(id)	Branch receiving the stock
requested_by	INTEGER	Foreign Key → users(id)	User who initiated the transfer request
approved_by	INTEGER	Foreign Key → users(id)	User who approved or rejected the transfer
status	VARCHAR(50)	Default 'pending'	Transfer status: pending, approved, or rejected
priority	VARCHAR(50)	Default 'normal'	Urgency level: normal or urgent
reason	TEXT	Nullable	Reason for requesting the transfer
notes	TEXT	Nullable	Additional operational notes

rejected_reason	TEXT	Nullable	Reason provided if the transfer was rejected
created_at	TIMESTAMP	Default NOW()	Timestamp when the transfer was requested
updated_at	TIMESTAMP	Default NOW()	Timestamp of the most recent status update

Table 7: Stock Transfers Table

Discrepancy logs table

Attribute Name	Data Type	Constraint	Description
id	SERIAL	Primary Key	Auto-increment unique identifier
product_id	INTEGER	Foreign Key → products(id)	The product with the detected discrepancy
branch_id	INTEGER	Foreign Key → branches(id)	The location where the discrepancy was detected

expected_quantity	INTEGER	Not Null	System-calculated expected stock level
actual_quantity	INTEGER	Not Null	Recorded or physically counted actual stock level
reason	TEXT	Nullable	Explanation or classification of the discrepancy
reported_by	INTEGER	Foreign Key → users(id)	User who flagged or confirmed the discrepancy
created_at	TIMESTAMP	Default NOW()	Timestamp when the discrepancy was logged

Table 8:Discrepancy Logs Table

Customers Table

Attribute Name	**Data Type**	**Constraint**	**Description**
id	SERIAL	Primary Key	Auto-increment unique identifier

name	VARCHAR(255)	Not Null	Full name of the customer or business entity
phone	VARCHAR(50)	Nullable	Customer contact phone number
email	VARCHAR(255)	Nullable	Customer email address
tin	VARCHAR(50)	Nullable	Rwanda Revenue Authority Tax Identification Number
tin_verified	BOOLEAN	Default FALSE	Indicates whether the TIN has been verified
address	TEXT	Nullable	Physical address of the customer
customer_type	VARCHAR(50)	Default 'retail'	Customer classification: retail or wholesale
credit_limit	DECIMAL(12,2)	Default 0	Maximum credit allowed for this customer in RWF
credit_used	DECIMAL(12,2)	Default 0	Current outstanding credit balance in RWF

notes	TEXT	Nullable	Additional notes or remarks about the customer
created_at	TIMESTAMP	Default NOW()	Timestamp when the customer record was created

Table 9: Customers Table

System Architecture Design

System architecture constitutes a conceptual framework that delineates the structural organization, operational characteristics, and analytical perspectives of a technological solution. It functions as a strategic blueprint, ensuring harmonious coordination among hardware infrastructure, software applications, and human operational resources (Pressman, 2014). For the StockSync Multi-Location Inventory Reconciliation System, the architectural design governs secure interactions across diverse user categories, including Administrators, Warehouse Managers, and Retail Cashiers.

To ensure optimal modularity, code reusability, and maintainability, StockSync is engineered utilizing a modern, decoupled Client-Server architectural pattern that strictly adheres to the Model-View-Controller (MVC) design paradigm. As illustrated in the architecture diagram below, this ecosystem is divided into three primary functional layers:

1. The Presentation / User Interface Layer (View)

This layer consists of the frontend client applications built utilizing React.js to ensure a highly dynamic, fast-loading user experience. It provides specialized interfaces tailored to user roles, such as the Administrative Dashboard for global multi-branch visualization and the Mobile Point-of-Sale (POS) View, which features rapid digital search interfaces and optional barcode scanning to accelerate retail checkout processing.

2. The Application / Logic Layer (Controller)

Operating as the intermediary conduit, the backend server is engineered using NodeJS and Express.js on a non-blocking architecture. This layer handles core business logic, processes incoming HTTP requests via secure RESTful Application Programming Interfaces (APIs) and manages the Automated Reconciliation Engine.

3. The Data Layer (Model)

The foundational layer is powered by a centralized PostgreSQL relational database. This system was specifically selected for its rigorous ACID (Atomicity, Consistency, Isolation, Durability) compliance, which is mandatory to prevent partial data writes and ensure data integrity during complex, multi-echelon stock transfers.

Operational Data Flow and Security

To interact with the StockSync platform, stakeholders establish connectivity through web browsers accessible via mobile devices or desktop computing platforms. Upon platform access, all user credentials are cryptographically hashed using the Bcrypt algorithm. The application server processes these authentication requests, utilizing JSON Web Tokens (JWT) to ensure strict, stateless authorization.

Once authenticated, the browser interface transmits specific operational requests (e.g., initiating a stock transfer) to the application server. The server coordinates with the PostgreSQL database to retrieve or modify pertinent inventory metrics. The database subsequently extracts the requested information and transmits the response back through the API to the frontend, delivering real-time processed data to the user. This architecture is compiled for production deployment via a Content Delivery Network (CDN) and scalable cloud hosting, utilizing Transport Layer Security (TLS/SSL) encryption to protect all commercial data in transit.

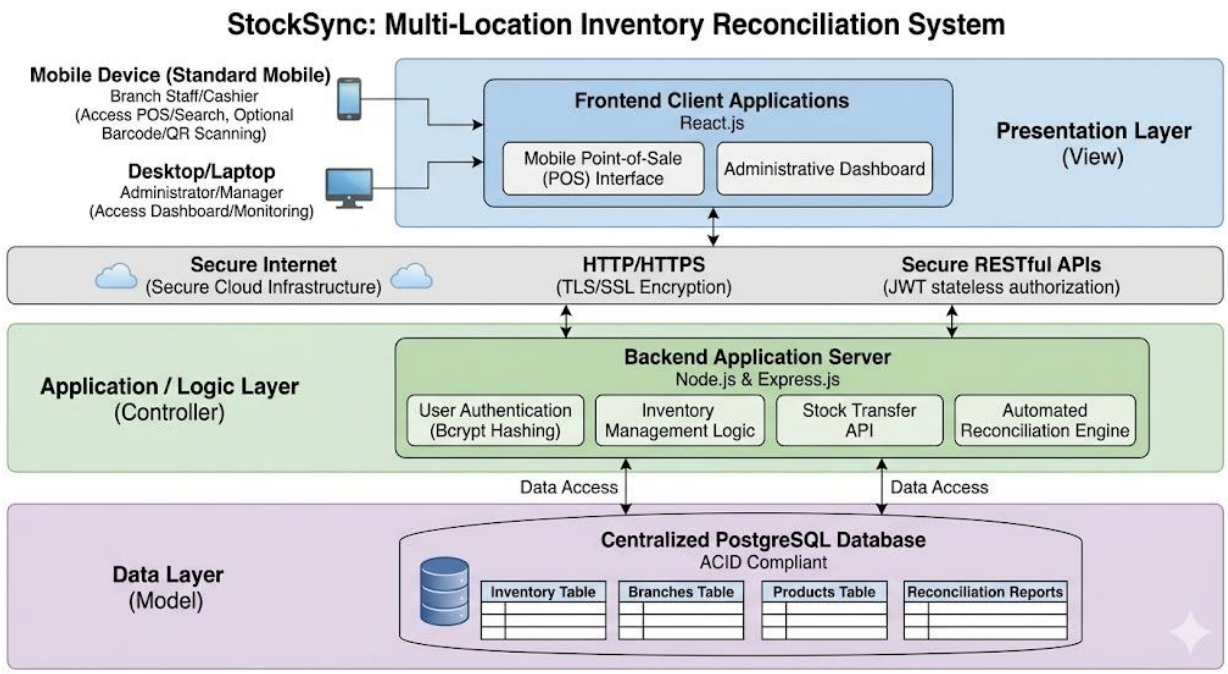

Figure 6: Architecture Design for the StockSync System

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